INFORMATION SYSTEMS

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THE UNIT EQUIPS LEANERS WITH BROAD KNOWLEDGE ON TYPES ,CHARACTERISTIC, FUNCTIONS OF INFORMATION SYSTEMS USED BY VARIOUS ORGANISATIONS

INFORMATION SYSTEMS

- An integrated set of components for collecting , storing ,processing data and providing information & digital products(knowledge)

COMPONENTS OF INFORMATION SYSTEM

A. HARDWARE

- These are the physical parts or features of an information system that enhances accurate operations of an information systems
- Monitors , keyboards , printers ,router , switch etc are some of system hardware

A. SOFTWARE

- These are the programs used to organize, process and analyze data
- Operating system , ultility programs , database are software that aid data procession

A. NETWORK

- Signals that facilitates operations of an information system
- Enhances accurate transmission , storage of data among information sys

COMPONENTS OF INFORMATION SYSTEM

a) PEOPLE

- ❖Agents that feds data into an information system
- ❖Enhances a proper data storage , procession (reports) and system operation

E. DATA

- ❖Unprocessed(raw) information fed to the system for procession
- ❖The information is used for making just decisions & strategic plans

TYPES (COMMON) INFORMATION SYSTEMS

a. TRANSECTION PROCESSING SYSTEM (TPS)

- An information system that captures organizational daily transections or operations
- Used by lower level managers to keep track of daily transections such as stock items , sales transections etc
- The system is mostly used by frontline or routine staffs i.e. bank tellers , till operators etc.

TYPES (COMMON) INFORMATION SYSTEMS

B. MANAGEMENT INFORMATION SYSTEMS (MIS)

- An information system comprising of summarized information or data
- Information is summarized in charts & diagrams such as pie charts ,tree diagrams etc.
- Used by mid-level or senior managers to determine the performance of organizational operations and making just(right) decisions
- Data in a management information system is derived from the transection processing system

TYPES (COMMON) INFORMATION SYSTEMS

C. DECISION SUPPORT SYSTEM (DSS)

- A knowledge based information system that facilitates formal or just decision making that affects organizational operations
- Helps in analyzing the existing structured information and project potential effects of decisions into the future
- Used to solve ill structured problems & enhances information exchange within the organization
- The system is used by senior managers

TYPES (COMMON) INFORMATION SYSTEMS

D. EXECUTIVE INFROMATION SYSTEMS (EIS)

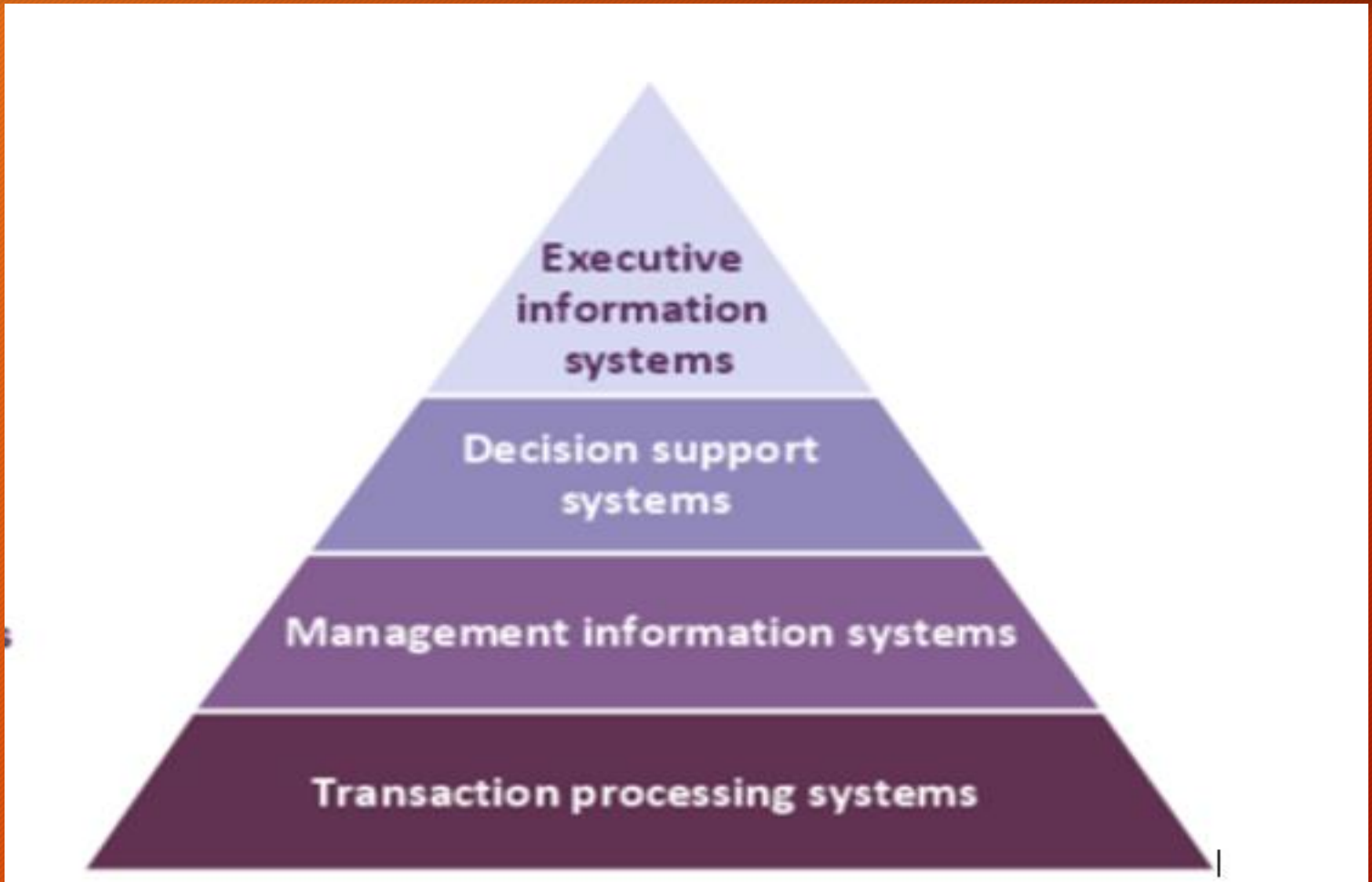
- An information system that helps in analyzing the environment in which the organization operates and identify long-term trends
- Used in making organizational strategic plan by the executives
- Comprises of weakly structured(undetailed)information which comes from both internal and external sources
- Information in an executive information system does not involve an intermediaries hence is directly represented by the executive

TYPES (COMMON) INFORMATION SYSTEMS

E. OFFICE AUTOMATIO

- system that enables data to move from one system to another without human interventions
- Allows organizational tasks to be accomplished faster
- Associate with small data storage volume
- Allows carrying out multiple tasks at once or simultaneously
- Reduces operational costs
- Electronic publishing ,electronic communication, image processing , office management

TYPES (COMMON) INFORMATION SYSTEMS



USES(IMPORTANCE) OF AN INFORMATION SYSTEM ON ORGANISATIONS

i. Facilitates clear communication as it comprises structured information that is directly understood by organizational key stalkeholders
ii. Enhances just decision making as the system comprises of summarized and up to date information that helps in choosing alternative course of action
iii. Enhances an accurate record keeping as the system consists of the database that stores data in tables
iv. Enhances accurate information sharing among key organization stakeholders through office automation

CHARACTERISTICS OF INFORMATION SYSTEM

A. AVAILABLE

- The system should be awalys available , consistent in performing its operations
- Consistent inform of facts and verifiable

B. COMPLETE

- The system should consists all features that facilitates accurate record keeping i.e. software, hardware , data etc.
- The system should be capable of performing its operations without much errors (faults)

CHARACTERISTICS OF INFORMATION SYSTEM

C. RELEVANT

- Data in the system should be meaningful hence its output should be efficiency in projecting organization decisions
- The system should be capable of functioning to its intended purpose i.e. a payroll management system should be dealing with organizational finances(salaries)

D. ACCESSIBILITY

- The system should comprise of all accessibility features i.e. physical challenged people
- Voice recognition for visual impaired users , joystick for the amputated users

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

Office applications are software's that automates (moves data within systems without human intervention) organizational business functions or processes

A. MICROSOFT WORD

- A word processing program used to create various official documents that can be printed or published
- The program comprises of various features to automate business functions or operations
- Mail merge is a feature used in producing multiple copies of a document at once
- Watermark a feature that enhances document security

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

B. MICROSOFT EXCEL

- A spread sheet program that organizes organizational data or information by presenting it in a grid of text (rows) & columns
- The program comprises of various formulas used in calculations namely: statistical formulas(sum , mean), logical & look up functions, pivot tables etc.
- the program summarizing data from large amount of information using pivot tables & charts
- Used in creating formal budget (monthly expenditure)
- Used in sorting & storing complex data

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

C. MICROSOFT ACCESS

- A database management system used to accurately store and manipulate organizational data
- Its consists of an interface , forms ,tables ,templates ,queries used to store, retrieve ,manage and analyze data effectively
- The program is used in producing accurate data or forensic reports

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

D. MICROSOFT POWER POINT

- A presentation program used to create slide shows composed of text and images(graphic)
- Enhances a clear communication of organization data as data is presented in slide shows & displayed on screen and printed

E. MICROSOFT PUBLISHER

- A Microsoft application used for designing brochures(book covers) , labels , calendars , business cards etc.

How office application are used by organizations

A. COMMUNICATION

- Mail merge in Microsoft word helps in efficiently creating one document & sending to thousands people with same address (header)
- Letter heads facilitates mail merge in Microsoft words while slide shows in Microsoft PowerPoint enhances an effective presentation of organizational data either on screen or printed documents

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

B. Office Management

- forms & advanced queries in Microsoft access are used to efficiently store and manipulate organizational data
- Pivot tables , charts ,statistical formulas in Microsoft excel are used to organize & summarize complex numerical data
- Letter head in Microsoft word is used in mail merging thousands documents thereby reducing human efforts

COMMON OFFICE APPLICATIONS (word processors , spreadsheet etc.)

C. COLLABORATION

- Mail merge and macros enhances collaboration of various office business processes i.e. financial reports , letters etc.

FUNCTIONS WITHIN AN ORGANISATION

- Functions are various tasks performed regularly by an organization
- Marketing, sales , distributes, accounting , logistics , management ,production , purchasing are examples of organizational functions

RANGE OF FUNCTIONS WITHIN AN ORGANISATION THAT INFORMATION SYSTEM SUPPORT

a) LOGISTICS

- An information system aids logistics through;
  - Routine planning and calculations
  - Dispatching of goods , costing and shading
  - Loading and delivery tracking

RANGE OF FUNCTIONS WITHIN AN ORGANISATION THAT INFORMATION SYSTEM SUPPORT

b) .MANAGEMENT

❖The system facilitates organizational management through accurate decisions and expenditure i.e.

1. Order Management

> eliminates human efforts as it gives access to customers to order products or goods online

> enhances a quick delivery of goods and services

RANGE OF FUNCTIONS WITHIN AN ORGANISATION THAT INFORMATION SYSTEM SUPPORT

2. Assert Management

- ❖Enhances a formal recording or tracking of organizational property
- ❖Property are recorded in system pivot tables , charts etc
- ❖Recorded data is outpted or generated through reports etc.

RANGE OF FUNCTIONS WITHIN AN ORGANISATION THAT INFORMATION SYSTEM SUPPORT

3. Communication

- Data and information are exchanges quickly through e-commerce business
- Improves customer services i.e. customer can order online ,check account status since most of the functions are automated

FUNCTIONALITY OF COMMON INFORMATION SYSTEMs

- Enhances a quick exchange of data or information as most of the features are automated i.e. AOS(automated Office system)
- Improves customer services i.e. customer can order or carry out their functions online i.e. account transections
- Eliminates human efforts as most operations are automatically or symiteniously carried out
- Helps in making concrete organizational decisions , strategic planning and budgeting i.e. decision support system
- Enhances accurate management of organizational business functions namely;
  - > an effective routine planning(scheduling of goods dispatch
  - > proper staff recruitment and training by the human resource department

BENEFITS OF COMPURISED INFORMATION SYSTEM OVER NON-COMPURISED SYSTEM

- A computerized system is a system that involves carrying out organizational business functions using an information system

BENEFITS

- Reduces costs as most of tasks are automated
- Data and information is exchanged quickly i.e. emails , reports etc.
- Customer services are improved i.e.an online transections
- Eliminates human efforts as most of the operations are carried out symeatenously i.e. meil merging

NOTE: Non-compurised system is a traditional system that involves carrying out organization functions through posts , keeping records in papers etc.

ROLES PREVIOUSLY MANAGED THROUGH PEOPLE OR PHYSICAL SYSTEM

- Keeping organizational records or data on paper files
- Sending organizational business contracts or processes through posts
- Communicating business transections through telephones , sms etc.

IDENTIFYING PEOPLE WITHIN AN ORGANISATION WITH KEY STAKES / MANAGEMENT RESPONSIBILITY FOR AN INFORMATION SYSTEM

a. SYSTEM SECURITY MANANGER

- An IT professional who protects information systems from breaches or attacks
- It could be cyber or physical attacks
- Implements system protection measures or standards such as intrusion detection system (strong firewalls , surveillance cameras), security system policies etc.

IDENTIFYING PEOPLE WITHIN AN ORGANISATION WITH KEY STAKES / MANAGEMENT RESPONSIBILITY FOR AN INFORMATION SYSTEM

b. DATABASE ADMINISTRATOR

- An IT professional who manages database system for an organization
- Creates and maintains organization database that are part of the system
- Protects organizational data by assigning CRUD (create ,retrieve ,update ,delete) operations to system users

IDENTIFYING PEOPLE WITHIN AN ORGANISATION WITH KEY STAKES / MANAGEMENT RESPONSIBILITY FOR AN INFORMATION SYSTEM

C. COORDINATOR

- Organizational employee who overseas dairy to dairy operations or business functions
- Overrides system security CRUD operations under routine staffs
- Ensures that the system outputs accurate information

IDENTIFYING PEOPLE WITHIN AN ORGANISATION WITH KEY STAKES / MANAGEMENT RESPONSIBILITY FOR AN INFORMATION SYSTEM

D. MANAGER

- Organizational senior officer who overseas all organizational functions
- Makes strategic organizational plans such as long and short term plans
- Makes concrete decisions affecting the organization hence associates much with an executive and decision support system

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

Data Protection

- The process or concept of protecting/ safeguarding organization information or data from thief, modification ,lost etc.
- Organizational executives , system security managers ,system analysts , system administrator are some of security personnel in protection of system data

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

A. SYSTEM SECURITY ANALYST

- An IT professional who interacts with the system during threats identification or detection
- Performs various threats regulatory programs i.e. antivirus installation ,system defragmentation

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

B. SYSTEM SECURITY MANAGER

- An IT professional who protects data by implementing security policy, intrusion detection system & data back up servers
- Audits system performance either monthly ,annually etc.

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

C. SYSTEM END USERS

- Organizational employee who carries out routine operations or business functions
- Carries out their operations according to their system CRUD functions namely: updating , deleting & retrieving data
- Encrypts packets(data transmitted over internet) and performs daily data back ups

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

D. DATABASE ADMINISTRATOR

- An IT personnel who manages database system for an organization
- Creates and maintains databases that are used to store organization data (information)
- Protects data by assigning CRUD operations to system users (employee)

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

E. ORGANISATION EXECUTIVES

- Managers , coordinators protect data by penalying employees in breach of the system security policies
- Ensures that end users abide to data protection measures such:
  - backing up data
  - Running antivirus programs from trusted sources
  - Not mounting any external drives to the system
  - Not clicking any link from untrusted sources
  - A hearing to CRUD operations

MANAGEMENT ROLES WITH RESPONSIBILIES FOR DATA AND DATA PROTECTION

F. SYSTEM TRAINER

- An IT professional who provides system security awareness to end users
- Sensitizes end users about how to properly operate the system , protect the system from threats etc.

IMPORTANCE OF AVOIDING A FRAGMENTED APPROACH TO INFORMATION MANAGEMENT

FRAGMENTATION

- A concept or condition by which a system main memory is broken down into smaller pieces after a process transmitted to the secondary memory
- The main memory bacames unused due its smaller size (a process in secondary memory can not be transmitted back to the main memory)

IMPORTANCE OF AVOIDING A FRAGMENTED APPROACH TO INFORMATION MANAGEMENT

DATA FRAGMENTATION

- A process in which data in memory is broken up into many pieces or duplicate (Random Access Memory[RAM] etc.)
- Fragmentation approach is an approach in which data is duplicated in many components of an information system i.e. hard disk drives

METHODS OF AVOIDING OR HANDLING DATA FRAGMENTATION

a) Metadata servers

- Servers that provides an accurate way of storing and managing information
- Involves storing data in specific fields thereby preventing data duplications

b) Disk Partition

- Enhances defragmentation that separates file system associated with more reads(frequently accessed)
- Allocates data into its format and storage location hence easy accessed

METHODS OF AVOIDING OR HANDLING DATA FRAGMENTATION

C. Clear Temporary files

- Frequently freeing system disk space i.e. recyle bin
- Uninstalling all useless system programs or softwares i.e. easy video maker etc.

D. Keep software or drivers updated

- System essential programs need to be updated frequently to leastest versions i.e. antivirus software

E. Keep File equal to block size

- Stores files in alternant to the volume of the storage capacity

METHODS OF AVOIDING OR HANDLING DATA FRAGMENTATION

F. Defrag hard drive regularly

- Disk defragmentation(process of reducing the degree of fragmentation) re-arrange and collect all file pieces thereby storing them in a single location of the disk
- Helps in freeing up more disk space and managing data effectively

RESPONSIBILITY OF INDIVIDUAL AND GROUPS FOR SECURITY

A. ORGANISATION EXECUTIVES

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RESPONSIBILITY OF INDIVIDUAL AND GROUPS FOR SECURITY

B. SYSTEM SECURITY MANAGER

- An IT professional who prevents system breaches by implementing security policy, intrusion detection system ,flood barriers etc.
- Detection system could be :
  - strong firewalls
  - Biometric access measures(fingerprints)
  - Surveillance cameras
  - Fire and smoke alarms
- Audits system performance either monthly ,annually etc.

RESPONSIBILITY OF INDIVIDUAL AND GROUPS FOR SECURITY

c. Technical Architects

- IT professionals who designs security solutions and performs system security tests
- Defines the overall system security structure and ensures that it functions at high standards

RESPONSIBILITY OF INDIVIDUAL AND GROUPS FOR SECURITY

D. SYSTEM END USERS

- Organizational employee who carries out routine operations or business functions
- Protects the system by:
  - Ruining antivirus programs dairy or hourly
  - Installing up to date software and from trusted sources i.e. antivirus software
  - Not mounting infected external drives or sticks to the system
  - Abiding to the system security policy

KEY POINT OF INTERACTION BETWEEN INFORMATION SYSTEMS AND PEOPLE

A. System Breaches

- An interaction is carried out with the system security manager during identifying breaches or attacks compromising the system
- An IT professional carries out various tasks to identify threats namely:
  - Ruining up to date antivirus software
  - Configuration of firewalls
  - penetration testing

KEY POINT OF INTERACTION BETWEEN INFORMATION SYSTEMS AND PEOPLE

B .Dairy routines

- An interaction is carried out with the frontline organization staffs when performing dairy operations i.e. entering of sales data
- Routine staffs are the most individuals associating with the system

KEY POINT OF INTERACTION BETWEEN INFORMATION SYSTEMS AND PEOPLE

c. Report Generation

- An interaction is made with the organizational coordinators or executives or staffs when generating reports
- It could be financial reports ,sales reports , school reports depending with the type of the system in use

KEY POINT OF INTERACTION BETWEEN INFORMATION SYSTEMS AND PEOPLE

D . Record Keeping and decision making

- An interaction is carried out when storing organizational information
- The information is used in making strategic plans for the company ,business etc. by the executives
- Information is kept in the system database

PHYSICAL & ELECTRONIC SECURITY CONCERNS(THREATS)ASSOCIATED WITH AN INFORMATION SYSTEM

Physical & electronic security concerns are events or measures that could compromise an effective operations of an information systems

- Electronic security events are castrated through the use of internet
- Physical security events are deliberate actions that compromise effective system operations

PHYSICAL & ELECTRONIC SECURITY CONCERNS(THREATS)ASSOCIATED WITH AN INFORMATION SYSTEM

Electronic security concern

a) Denial of Service Attack(DOS)

- ❑An attack that shuts down a system a system or network ,making it inaccessible to its intended users (employee, customers etc.)
- ❑The attack highly targets the web servers of high profile organization such as banking , commerce & media companies
- ❑The attack leads to loss of significant information & money
- ❑A flooding services or crushing services is DOS attack where the system receives too much traffic for the server to buffer ,thereby slowing down the system performance
- ➢Network traffic is amount of data moving across a network at a given point of time

PHYSICAL & ELECTRONIC SECURITY CONCERNS(THREATS)ASSOCIATED WITH AN INFORMATION SYSTEM

B. **Social Engineering**

- an attack which occurs through the use of social sites
- Attackers creates false social sites to obtain organizational confidential details
- Prompts employee to log in the ghost social site account with organizational credentials

PHYSICAL & ELECTRONIC SECURITY CONCERNS(THREATS)ASSOCIATED WITH AN INFORMATION SYSTEM

C. **Cross Site Script**

- a threat in which an attacker injects high traffic to client site script in order to buy pass access controls
- Such sites when run , an attacker obtains / gets the access controls
- The threat is much applied / associated with web applications

RISK MANAGEMENT STRATEGIES

RISK MANAGEMENT

- risk management is the process of managing information system risks by identifying , assessing & controlling risks
- Risks management helps to control the uncertanity

RISK MANAGEMENT STRATEGIES

SYSTEM RISK

- Uncertain event or threat that could compromise normal operations of an information system
- Human errors , natural disaster, electronic faults are measures that brings about risks
- System crash or corrupt is an example of an information system risk

RISK STRATEGIES

- Measures undertaken to control information system uncertainty
- Risk avoidance ,acceptance ,mitigation and transfer are strategies used in handling the uncertainty

A. RISK ACCEPTANCE

- A strategy that involves acknowledging the risk
- Involves taking no action until it occurs or happen

RISK STRATEGIES

B. RISK AVOIDANCE

- A strategy that involves preventing the risk from happening or occurring
- Requires eliminating the threat and protect the system from its impact

RISK STRATEGIES

C. RISK MITIGATION

- A strategy that involves decreasing the probability of risk occurrence or impact of the threat
- Involves undertaking measures such as:
  - Frequently backing up system data
  - Frequently performing penetration testing
  - Frequently updating & running antivirus softwares

RISK STRATEGIES

D. RISK TRANSFER

- A strategy that involves giving the responsibility of handling the threat or impact to another party
- The risk could be transferred to a risk handler , insurance company etc.

KEY ASPECT OF SECURITY POLICIES(CONFIDENTIALITY ,ACCESS,ACCOUNTABILITY , AUNTHENTICATION ETC)

IT SECURITY POLICY

- A document comprising of set of rules , standards enacted to prevent data and information system breaches
- Guides system end users on how to effectively protect the system from threats
- The policy need to be updated frequently and enforce data confidentiality , integrity(accountability) , availability , authentication etc.

KEY ASPECT OF SECURITY POLICIES(CONFIDENTIALITY ,ACCESS,ACCOUNTABILITY , AUNTHENTICATION

A . CONFIDENTIALITY

- An aspect of protecting sensitive system data from unauthorized access
- Involves encrypting data in storage and over transmission

KEY ASPECT OF SECURITY POLICIES(CONFIDENTIALITY ,ACCESS,ACCOUNTABILITY , AUNTHENTICATION

B. INTEGRITY

- An aspect that involves protecting data from unauthorized deletion or modification
- Involves assigning CRUD operations to system users according tp their roles in the system
- Others could be creating entries , deleting , updating and retrieving

KEY ASPECT OF SECURITY POLICIES(CONFIDENTIALITY ,ACCESS,ACCOUNTABILITY , AUNTHENTICATION

C. AVAILABILITY

- An aspects ensuring that there is uninterrupted access to an information system to authorized users
- Involves implementing robust servers ,strong firewalls , uninterrupted power supply(UPS) etc.

D. Authentication

- An aspect that checks if the system user is legible to access the system
- Involves checking login credentials such as passwords
- Involves creating strong passwords

LEGAL OBLIGATIONS OF INFORMATION SYSTEMS PROFESSIONALS

Legal Obligations

- These are standards set forth in government laws to protect information systems by various professionals
- ISO 27001 is of the legal oblication that protects an information system
- ISO 27001 is an international standard that allows or edges system professionals to undertake the foolowing operations:

ISO 27001 legal obligation

1. System security manager

- An IT professional who protects an information system threats or breaches either physical or cyber attacks
- Carries out penetration testing (a test done to examine loopholes or weakness in the system)
- Frequently maintenances the system either hardware , network and software architecture
- Defines a system security policy and frequently updates it

ISO 27001 legal obligation

2. System Analyst

- An IT professional who understands in details information system problems & solves them technologically
- Ensures system users not to create weak system passwords rather strong passwords comprising of letters , symbols and numeric
- Prevents system users from shareling system passwords
- Carries out compliance auditing (a process of evaluating whether system users abits to system security policy)

ISO 27001 legal obligation

3. System trainer

- An IT professional that sensitizes and trains system users on the use of the new system
- Allows the training to communicated through informal and formal communication (sign languages for the physically challenged users)
- Allows to define strategic system training guides i.e. focusing on impacts of cyber attacks & its mitigation strategy , sabotage and its mitigation strategy)

ISO 27001 legal obligation

4.Database administrator

- An IT professional who manages an organizational database system
- Assigns accurate system access rights to user called CRUD operations where some users could be creating data , Retrieving data , Updating data and Deleting data.
- frequently encrypting data in storage and over transmission using standard encrypting tools (MD5)
- Frequently monitors the database system and backs up data

ISO 27001 legal obligation

Network Administrator

- An IT professional who overseas the network of an organizational information system
- Monitors and controls network or cyber threats using network map (NMAP) , wireshark etc.
- Configures strong system firewalls and network architectures (star topology , hybrid topology etc.)